

TUSHARIKA RASTOGI

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PROFESSIONAL SUMMARY

Computational Biologist specializing in cancer genomics and multi-omics analysis. Expertise spans variant calling, CNV analysis, mutational signature profiling, and RNA-seq, executed on AWS Batch and HPC using Nextflow. Builds production-grade pipelines and custom automation tools to improve reproducibility and reduce manual review across cohorts. Incorporates AI-assisted development into daily workflows. Independent, async-native, and experienced translating complex genomic analyses for clinical and scientific stakeholders across distributed teams.

EXPERIENCE

Michigan Medicine **Ann Arbor, MI**
Bioinformatics - Computational Biologist May 2023 - Present

- Configured, scaled, and executed nf-core Nextflow workflows (Sarek for DNA variant calling, RNA-seq, and fusion detection) on AWS Batch and HPC, reducing processing time by 60–80% through optimized parallelization and resource configuration, while improving reproducibility.
- Designed and executed differential expression and pathway analysis pipelines across NSCLC cohorts (ALK, EGFR, KRAS) as primary analyst for a collaboration with Tempus AI.
- Identified clinically actionable variants and characterized mutational landscapes across 200+ tumor samples (PDX, patient-derived, and cell lines) using GATK Best Practices and VarSeq annotation, supporting precision oncology efforts across multiple active research cohorts.
- Performed copy number variation (CNV) analysis using Sequenza on 100+ tumor samples, with and without matched normal controls, inferring chromosomal alterations, tumor purity, and ploidy.
- Evaluated proteomics platform suitability for tumor sample analysis using limma-based differential abundance and dynamic range assessment.
- Characterized mutational signatures in tumor samples using SigProfiler, distinguishing endogenous vs. exogenous processes to uncover potential carcinogenic mechanisms.
- Developed internal automation utilities (e.g., dbNSFP Aggregator, cohort comparison metrics, annotated oncoplots) to reduce manual review time by over 50% and improve consistency across analyses, supporting interpretation by wet-lab and clinical collaborators.

Indian Institute of Technology **Gandhinagar, India**
Summer Intern May 2022 – July 2022

- Developed an R pipeline to analyze HNSCC transcriptomic data, identifying 895 DEGs and 69 DEMs.
- Revealed post-transcriptional regulatory mechanisms and candidate therapeutic targets through DEG–miRNA correlation and pathway enrichment analysis.

PUBLICATIONS & PRESENTATIONS

- Manuscript in Preparation - Brain Metastasis Cohort 2026
- Developing diverse patient-derived xenograft models of common and rare brain metastases to elucidate molecular landscapes and reveal therapeutic opportunities
2nd Author | Poster | [AACR Special Conference in Cancer Research: Brain Cancer](#) March 2026
- Examining Molecular Features of Rare Brain Metastases Patient-Derived Xenograft Models
4th Author | Poster | [Journal of the National Comprehensive Cancer Network](#) March 2025
- Examining molecular landscape and drug sensitivity profiles of patient-derived xenografts from resected brain metastases
2nd Author | Poster | [Journal of Clinical Oncology Volume 42](#) August 2024

EDUCATION

University of Michigan, Ann Arbor, MI - MS in Bioinformatics (GPA: 3.65/4.00) December 2022
Amity University, Noida, U.P., India - B.Tech in Bioinformatics (GPA: 4.0/4.0) June 2021

SKILLS

Programming & Infrastructure: R, Python, Shell Scripting, Nextflow, AWS (S3, Batch, EC2), HPC, Seqera Platform, Git, GitHub, Docker

Bioinformatics Expertise: DNA variant calling, bulk RNA-seq, scRNA-seq, CNV analysis, proteomics, mutational signature profiling, pathway enrichment, multi-omics integration, Next-Generation Sequencing Analysis

Statistical & Visualization Tools: DESeq2, Limma, Seurat, Maftools, ggplot2, ComplexHeatmap, Plotly, EnhancedVolcano, pandas, matplotlib, SciPy

Databases & Resources: TCGA, GDC, COSMIC, ClinVar, KEGG, MSigDB, gnomAD, 1000 Genomes, GEO, SRA, Ensembl, UniProt

Variant Interpretation & Data Formats: VarSeq, dbNSFP, VCF, MAF, BAM/CRAM, FASTQ

AI-Assisted Development: Claude, Claude Code, GitHub Copilot